23/4/2026

Recall

Rectification means finding lengths.

Cartesian coordinates

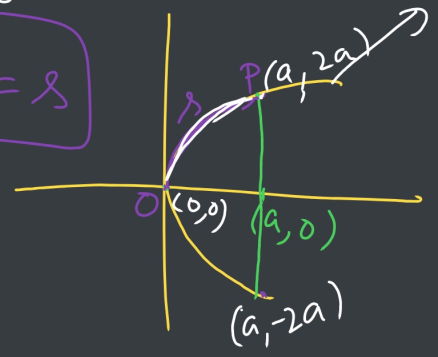
$$s = \begin{cases} \displaystyle\int_{x_1}^{x_2} \sqrt{1+\left(\frac{dy}{dx}\right)^2}\, dx \\ \displaystyle\int_{y_1}^{y_2} \sqrt{1+\left(\frac{dx}{dy}\right)^2}\, dy \end{cases}$$

① S.T the length of the parabola $y^2 = 4ax$ from the vertex to the end of the Latus rectum is $a\left[\sqrt{2} + \log(1+\sqrt{2})\right]$

Latus Rectum: Chord passing through the focus $(a,0)$

Find the length of the arc cut off by the line $3y = 8x$

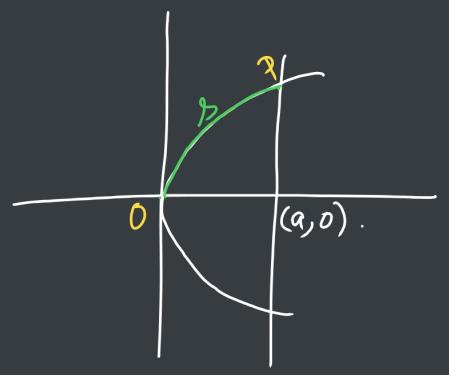
$\boxed{\text{To find } \widehat{OP} = s}$



$y^2 = 4ax$

$y^2 = 4a(a)$

$y = \pm 2a$



Reqd to find $s = \widehat{OP}$

The Latus Rectum is $\boxed{x = a}$

To find the Coordinates of points of intersection of LR with the parabola

We put $x = a$ in the eq^{n} of the parabola

($\because$ the intersection pts satisfy the eq^{ns} of both LR & parabola)

$$\therefore \quad y^2 = 4ax$$
$$= 4a(a)$$
$$\Rightarrow y^2 = 4a^2$$
$$\Rightarrow y = \pm 2a$$

Let's consider the first quadrant: & so we take $y = 2a$.

$\therefore \quad P = P(a, 2a)$

$\therefore \quad s = \widehat{OP}$

$\because (0,0)$ to $(a, 2a)$

Now

$$y^2 = 4ax$$
$$\Rightarrow x = \frac{y^2}{4a}$$
$$\Rightarrow \frac{dx}{dy} = \frac{2y}{4a}$$

ie $\dfrac{dx}{dy} = \dfrac{y}{2a}$

$$\because\ 1+\left(\frac{dx}{dy}\right)^2 = 1+\frac{y^2}{4a^2}$$

$$= \frac{4a^2+y^2}{4a^2}$$

$$\therefore\ s = \widehat{OP} = \int_{y=0}^{y=2a} \sqrt{1+\left(\frac{dx}{dy}\right)^2}\, dy$$

$(0,0)$ to $(a,2a)$ $y=0$

$$= \int_0^{2a} \sqrt{\frac{4a^2+y^2}{4a^2}}\, dy$$

$$= \frac{1}{2a}\int_0^{2a} \sqrt{4a^2+y^2}\, dy$$

$$\left[\because \int \sqrt{a^2+x^2}\, dx = \frac{x}{2}\left[\sqrt{a^2+x^2}\right] + \frac{a^2}{2}\left[\ln\left[\sqrt{a^2+x^2}+x\right]\right]\right]$$

$$\Rightarrow s = \frac{1}{2a}\left[\frac{y}{2}\left[\sqrt{4a^2+y^2}\right] + \frac{4a^2}{2}\left[\ln\left[\sqrt{4a^2+y^2}+y\right]\right]\right]_0^{2a}$$

$$= \frac{1}{2a}\left[\frac{\cancel{2}a}{\cancel{2}}\left[\sqrt{4a^2+(2a)^2}\right] + \frac{\cancel{4}^{2}a^2}{\cancel{2}}\left[\ln\left(\sqrt{4a^2+(2a)^2}+2a\right)\right] - \left[0 + \frac{\cancel{4}^{2}a^2}{\cancel{2}}\left[\ln\sqrt{4a^2+0}+0\right]\right]\right]$$

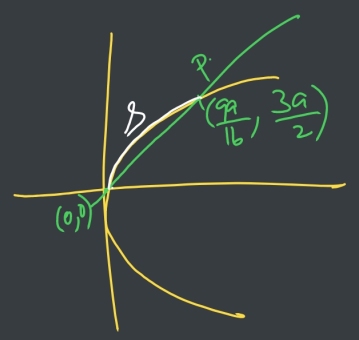
$$\Rightarrow s = \frac{1}{2a}\left[a\left[\sqrt{8a^2}\right] + 2a^2\left[\ln\left(\sqrt{8a^2}+2a\right)\right] - 2a^2\left[\ln(2a)\right]\right]$$

$$\Rightarrow s = \frac{1}{2a}\left[a\left[2\sqrt{2}\,a\right] + 2a^2\left[\ln\left(\frac{2\sqrt{2}\,a+2a}{2a}\right)\right]\right]$$

$$\Rightarrow s = \frac{1}{2a}\left[2\sqrt{2}\,a^2 + 2a^2\left[\ln\left(\frac{\cancel{2a}(\sqrt{2}+1)}{\cancel{2a}}\right)\right]\right]$$

$$\Rightarrow s = \frac{1}{2a}\left[2\sqrt{2}\,a^2 + 2a^2\ln(\sqrt{2}+1)\right]$$

$$\Rightarrow s = \sqrt{2}\,a + a\ln(\sqrt{2}+1) \Rightarrow \boxed{s = a\left(\sqrt{2}+\ln(\sqrt{2}+1)\right)}$$



Here the reqd arc length

$s = \overset{\frown}{OP}$

$(0,0)$ to $\left(\frac{9a}{16}, \frac{3a}{2}\right)$

line : $3y = 8x$

$\Rightarrow x = \frac{3y}{8}$

$y^2 = 4ax$

$\Rightarrow y^{\cancel{2}} = 4a\left(\frac{3\cancel{y}}{8}\right)$

$\Rightarrow y = \frac{3a}{2}$

$\Rightarrow x = \frac{3 \times \frac{3a}{2}}{8}$

$\Rightarrow x = \frac{9a}{16}$

$\therefore$ Here

$$s = \int_{y=0}^{\frac{3a}{2}} \sqrt{1+\left(\frac{dx}{dy}\right)^2}\, dy$$

$$\Rightarrow s = \int_0^{\frac{3a}{2}} \sqrt{\frac{4a^2+y^2}{4a^2}}\, dy = \int_0^{\frac{3a}{2}} \frac{1}{2a}\sqrt{4a^2+y^2}\, dy$$

$$\Rightarrow s = \frac{1}{2a}\left[\frac{y}{2}\sqrt{4a^2+y^2} + \frac{4a^2}{2}\left[\ln\left[\sqrt{4a^2+y^2}+y\right]\right]\right]_0^{\frac{3a}{2}}$$

$$= \frac{1}{2a}\left[\frac{3a}{4}\sqrt{4a^2+\frac{9a^2}{4}} + 2a^2\left[\ln\left[\sqrt{4a^2+\frac{9a^2}{4}}+\frac{3a}{2}\right]\right]\right.$$
$$\left. - \left[0 + 2a^2 \ln\left(\sqrt{4a^2}\right)\right]\right]$$

$$\Rightarrow s = \frac{1}{2a}\left[\frac{3a}{4} \times \frac{5a}{2} + 2a^2 \ln\left(\frac{\frac{5a}{2}+\frac{3a}{2}}{2a}\right)\right]$$

$$\Rightarrow s = \frac{1}{2a}\left[\frac{15a^2}{8} + 2a^2 \ln 2\right] = \frac{15a^{\cancel{2}}}{2\cancel{a} \times 8} + \frac{\cancel{2}a^{\cancel{2}}}{\cancel{2a}} \ln 2$$

$$\Rightarrow \boxed{s = a\left[\frac{15}{16} + \ln 2\right]}$$